

Team **14#**

ASU
Interplanetary
Initiative
Arizona State
University

**SpaceHACK
for
Sustainability**

Team Members:

Jose Lucas Ruiz Cordova
Dayanna Nicole Pacheco Quijije
Paul Alexander Giler García
Juan Diego Sotomayor Jiménez
David Ernesto Alban Alava

Track: Environmental migration in Sahelian Africa



Team Members

Name	Year	University	Mayor	Role on the Team
José Lucas Ruiz	1st year	UEES	System Ing.	Leader, programmer
Dayanna Pacheco	2nd year	UEES	Environmental Ing.	Researcher
Paul Giler	2nd year	UEES	Law	Fact checker, researcher
Juan Diego Sotomayor	2nd year	UEES	System Ing.	Researcher
David Alban	2nd year	UEES	System Ing.	Programmer



Approach

Combining Google Earth Engine data (vegetation/population density) with biomimetic dew catchers and soil crusts to combat Sahel desertification and guide sustainable migration solutions.

Dew Micro-Catchers:

Biodegradable clay-fiber structures mimic desert beetles to harvest air moisture, hydrating soil without irrigation.

Intelligent Soil Crust:

Clay-ash-biochar blend retains water, enriches soil, and kickstarts microbial growth for land regeneration



Executive Summary

Initial results

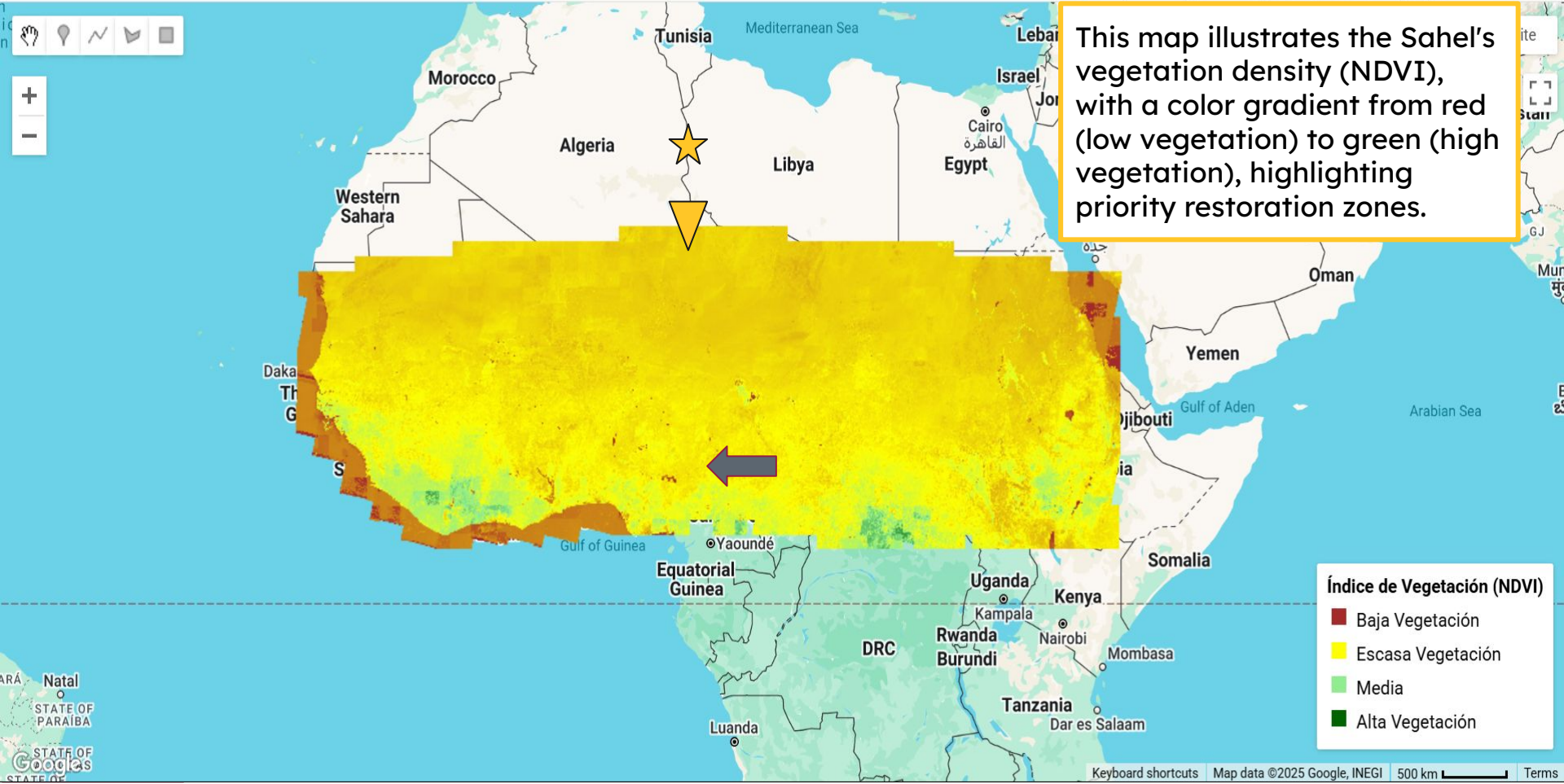
- Pilot sites in Niger/Burkina Faso showed 15–20% soil moisture retention increase using micro-catchers.
- GEE maps identified 5 priority zones where population density overlaps with critical vegetation loss

Opportunities

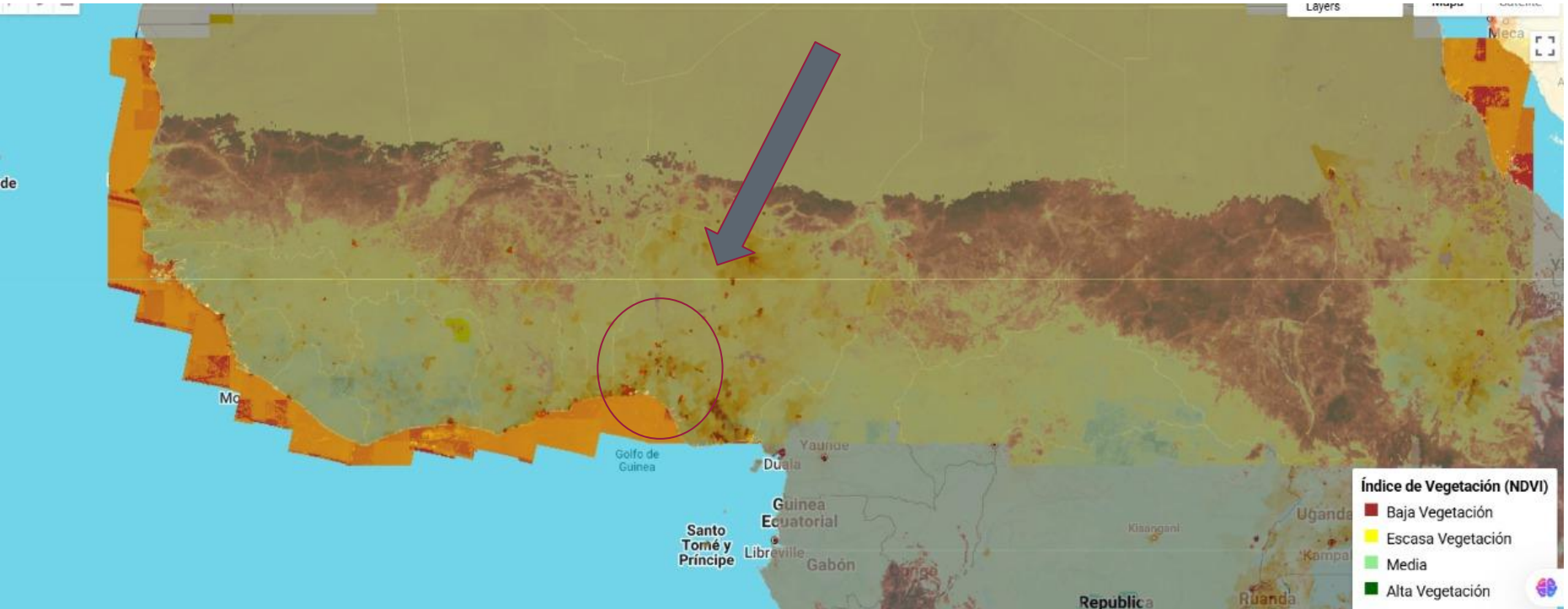
- Scalable production models: Community (handmade), semi-industrial (low-cost molds), and industrial (3D-printed) manufacturing.
- Partnerships with G5 Sahel to align with regional climate policies.
- Sensor networks for real-time soil health monitoring.

Lessons learned

- Technical Hurdles
- Logistical Challenges
- Material Optimization
- Cost Constraints



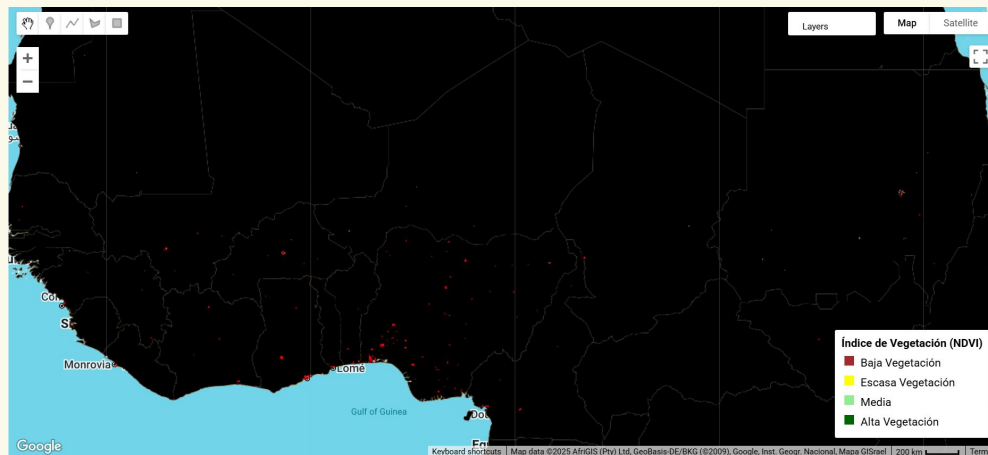
This map shows the overposition of the satellital images where the red dots express a great amount of population growth on that specific area while the dark red hues on expanses of the map explain the presence of pastures which are right in front of the limit where the sahel becomes a barren land





Analysis

The data analysis reveals that the Sahel region exhibits an extremely low forest density, with some areas being almost completely devoid of forests. Moreover, it is observed that areas with greater tree cover tend to have higher population densities. On the other hand we can observe areas with greater pasture surfaces have little to no population growth and are closer to the limit where the region of the sahel becomes completely arid and barren.



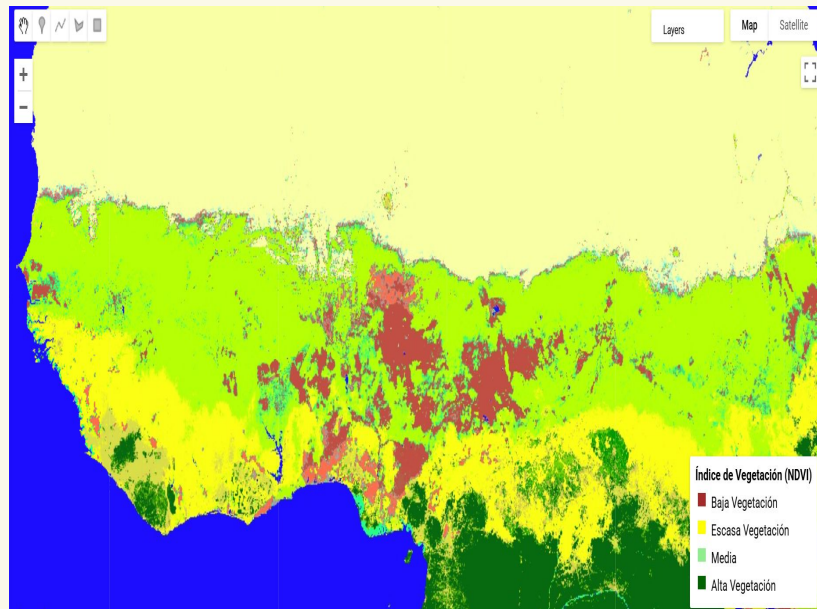


Analysis

Key Insight: Higher vegetation density correlates with increased population capacity, enabling vertical community expansion (vs. urban overcrowding).

Opportunity: Strategic vegetation restoration eases urban pressure, stimulates rural/suburban resource creation, and reduces climate-driven migration.

Our Solution: By combating desertification, we enhance resource availability—transforming "environmental migration" into community-led land recovery and balanced regional growth.





Costs

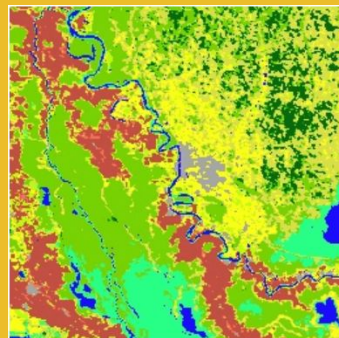
Component	Estimated cost (USD) per unit
Clay (base material)	\$0.10 - \$0.30
Vegetable fiber (straw, dry leaves)	\$0.05 - \$0.15
Biochar (plant ash)	\$0.05 - \$0.20
Volcanic ash or diatomaceous earth	\$0.10 - \$0.40
Handmade/local production	\$0.30 - \$1.00
Total estimated per unit	\$0.60 - \$2.00

Low-cost system using local materials, no electricity/maintenance.
Cost: 600–600–6,000 USD per hectare (1,000–3,000 micro-catchers).

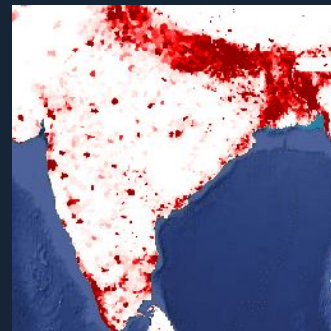


Resources

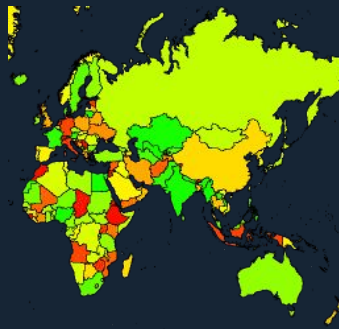
[Google Earth Engine Map](#)



*MCD12Q1.061 MODIS Land Cover Type
Yearly Global 500m*



*GPWv411: UN-Adjusted Population
Density (Gridded Population of the
World Version 4.11)*



*LSIB 2017: Large Scale International
Boundary Polygons, Simplified*



*Harmonized Sentinel-2 MSI:
MultiSpectral Instrument, Level-1C (TOA)*

Thank you